SMART CANTEEN MANAGEMENT SYSTEM

DESIGN MANUAL

Group 01

Table of Contents

Introduction.....02

Product Development.....04

System Overview.....07

Technology stack.....10

Hardware Implementation.....12

Mobile Application.....15

Pages and features.....20

Database.....29

Server.....30

Security and Privacy.....31

Testing.....33

Maintenance and Updates.....34

Conclusion.....36

Appendices.....37

Introduction

In an age where digital innovation shapes every aspect of daily life, the demand for efficient and intelligent systems within public and institutional services continues to rise. The **Smart Canteen Management System** stands at the forefront of this transformation—designed to streamline food service operations, enhance user convenience, and ensure transparency in campus dining facilities. With a focus on speed, hygiene, accessibility, and smart payment solutions, this system is poised to revolutionize the way students and staff experience campus dining.

Canteen Management Concerns

Traditional canteen systems often face numerous challenges, including long queues, cash handling issues, lack of transparency in billing, and miscommunication between users and merchants. These setbacks not only cause dissatisfaction but also result in operational inefficiencies. The Smart Canteen Management System addresses these pain points with a comprehensive digital solution, enhancing both service delivery and user experience.

User Expectations

Students and staff expect more than just a place to eat—they demand a modern, efficient, and user-friendly dining experience. From viewing daily menus to tracking expenses and reducing waiting time, users today seek seamless integration of technology into their mealtime routines. The Smart Canteen Management System meets these expectations with real-time queue detection, fingerprint/RFID authentication, online ordering, and budget tracking tools, delivering a personalized and efficient experience.

Technological Advancement

Incorporating cutting-edge technologies such as fingerprint authentication, RFID-based payments, real-time congestion analysis using ESP32 cameras, and intuitive mobile and web applications, the Smart Canteen Management System leverages innovation to offer a smart, data-driven solution. The system is designed to scale with evolving user needs while ensuring reliability and performance in high-demand environments.

Legal and Operational Compliance

The system complies with institutional standards for digital payment security, user data protection, and accessibility. It promotes cashless transactions, reduces human errors, and ensures accountability in every transaction, making it a robust solution aligned with modern operational best practices in institutional food service.

Setbacks in Current System

Conventional canteen management often relies on manual processes, leading to inefficiencies, fraud, lack of data analytics, and poor user satisfaction. With no mechanism to monitor demand or manage rush hours effectively, the existing systems fall short of delivering optimal service. The Smart Canteen Management System addresses these shortcomings by integrating automation, intelligent analytics, and real-time service updates.

Smart Canteen Management System

The Smart Canteen Management System is more than just an upgrade—it's a complete reimagination of how institutional food services can operate in the digital age. It brings together powerful backend management with sleek, user-friendly interfaces to offer a unified experience for users, merchants, and administrators.

Features

Queue Detection: Detect real-time congestion using ESP32 camera modules, helping users avoid peak times.
Digital Authentication: Fingerprint and RFID-based user identification for secure access.
Cashless Payments: Integrated RFID-based payment system, removing the need for physical cash.
Menu Management: Merchants can update the daily menu through a dedicated mobile app.
Real-Time Order Syncing: Orders are instantly visible to merchants; completed orders are removed to avoid duplication.
User Interfaces: Custom-tailored mobile apps for users and merchants, and a central dashboard for administrators.

The **Smart Canteen Management System** isn’t just a convenience—it’s a commitment to efficiency, hygiene, transparency, and user empowerment. As institutions strive for smarter campuses, this system sets a new standard in canteen operations, ensuring that every meal is served with speed, safety, and satisfaction.

Product Development

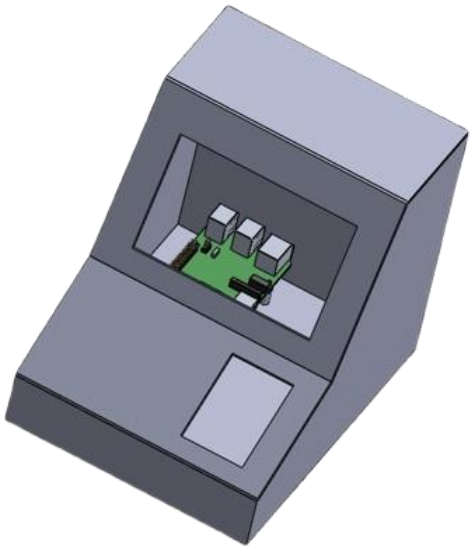
In building the hardware foundation of our **Smart Canteen Management System**, we thoughtfully designed **two dedicated casings**, each tailored to serve distinct functions while ensuring practical usability, robustness, and seamless interaction. These devices were engineered with real-world usage in mind, particularly focusing on the needs of both users and operators within a dynamic campus canteen environment.

Model Designs

We developed **two separate hardware modules**:

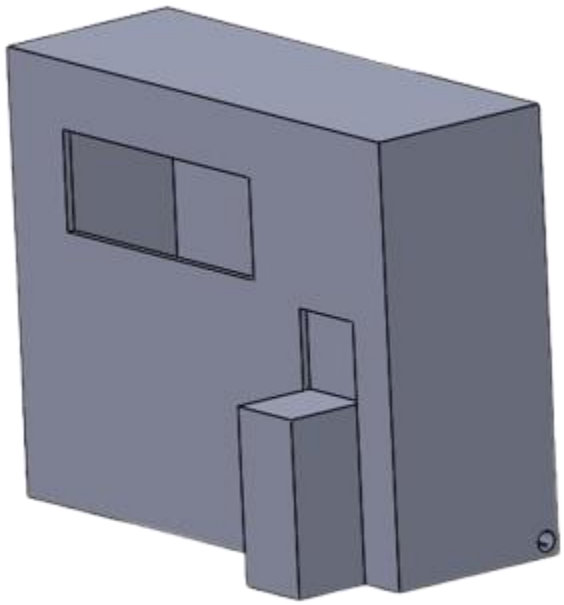
1. Primary Interface Module

- This unit houses the **Raspberry Pi module**, a **touch screen display**, and an **RFID reader**.
- The device is placed at user-accessible counters, allowing students to log in, view menus, place orders, and make payments via RFID.
- The **touch screen** was carefully embedded at an ergonomic angle to facilitate smooth navigation through the digital interface.
- The **RFID reader** was mounted on the lower front face of the unit to allow effortless tapping of RFID cards or tags during transactions.
- The ports were neatly routed to the rear panel to allow secure cable management and minimize visible clutter.



2. Biometric & Feedback Module

- This compact module integrates the **fingerprint scanner**, **LCD display**, and an **alert buzzer**.
- Designed for fast user authentication, this device is installed in areas where secure identification is required before proceeding with payment or order confirmation.
- The **buzzer**, placed directly beneath the LCD, provides instant audio feedback during fingerprint registration or scanning errors.



Typography

User interaction with both devices relies heavily on clear, real-time feedback. We selected **large, legible fonts** with high contrast for the LCD and touchscreen displays. Whether it's a successful registration, order total, or balance update, every message is presented in a way that is immediately readable, ensuring minimal confusion and maximum user confidence.

Final Models

The **final designs of both hardware casings** reflect our focus on modularity, clarity, and function.

- The **Raspberry Pi and touchscreen unit** forms the central hub of user interaction, offering a fluid digital experience backed by stable hardware and a responsive interface.
- The **biometric module**, meanwhile, is compact yet powerful—built to deliver reliable identification while keeping the user informed through visual and auditory cues.

Additional thoughtful touches include:

- **Ventilation grills** to protect internal electronics while allowing airflow.
- **Dedicated cable routing** paths to reduce tangling and improve aesthetics.
- **Rear-access panels** for maintenance and system upgrades.
- **Secure mounting slots** for both wall-mounted and tabletop installations.

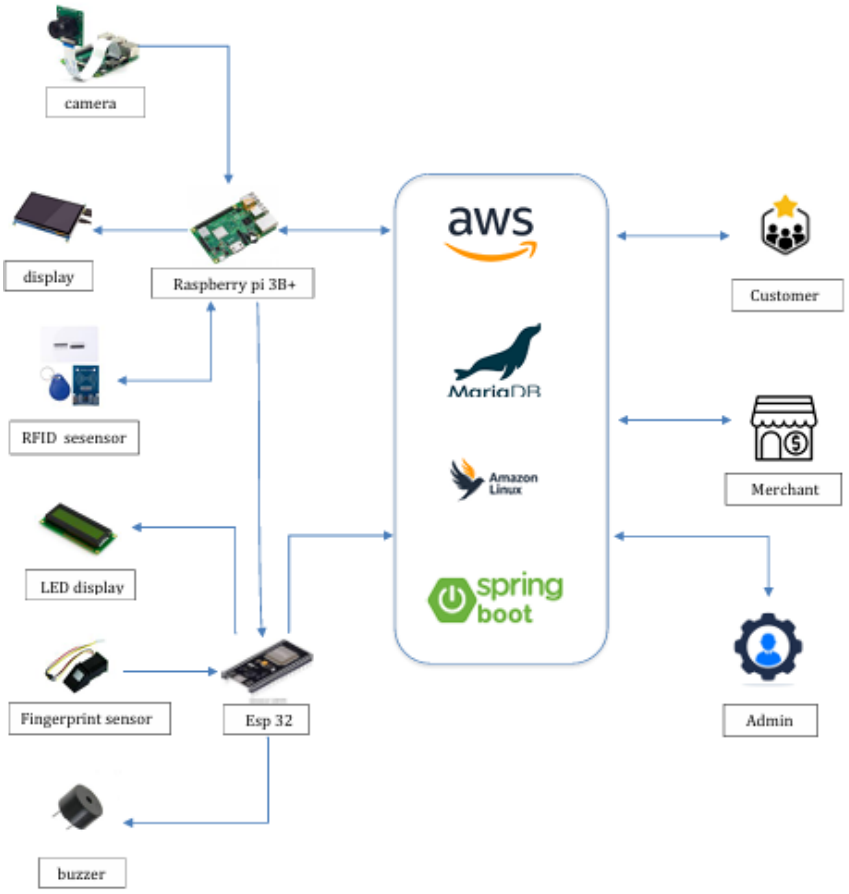
Together, these two devices create a **unified smart system**, enhancing canteen operations by automating authentication, enabling cashless transactions, and reducing queues—all while maintaining a high level of user-friendliness and operational stability. The design seamlessly bridges functionality and form, staying true to the goals of the Smart Canteen Management System.

System Overview

The **Smart Canteen Management System** is a complete end-to-end solution developed to modernize and optimize food service operations within campus environments. Designed primarily for university and school settings, the system automates the ordering, authentication, payment, and queue management processes—providing a faster, more hygienic, and user-friendly experience for students, merchants, and administrators.

The system integrates multiple hardware and software components including **RFID readers** for contactless payments, **fingerprint scanners** for biometric authentication, **ESP32 camera modules** for real-time queue monitoring, **LCD displays** for transaction feedback, and **buzzers** for status alerts. Together, these elements form a unified ecosystem that enhances transparency, improves efficiency, and reduces congestion in high-traffic canteen areas.

Solution Architecture



The solution architecture of the Smart Canteen Management System is thoughtfully designed to ensure **real-time responsiveness, modular scalability, and secure communication** between hardware, backend services, and user interfaces.

Core Modules & Flow

At the heart of the system are **two hardware units**:

- A **touchscreen-based Raspberry Pi module** that handles food item display, RFID-based payments, and user interaction.
- A **biometric terminal** equipped with a fingerprint scanner, LCD, and buzzer for authentication and confirmation.

These devices interact with a **Spring Boot–based backend**, hosted on a secure server environment. Data is transferred seamlessly through **HTTP APIs** to support functions like user registration, fingerprint/RFID mapping, order placement, balance deduction, and food item updates.

Key Technologies

- **ESP32 Camera Module**: Monitors live queues and sends periodic congestion data to the backend, allowing users to view congestion levels before arriving.
- **mariaDB**: Acts as the central database, storing user profiles, fingerprint hashes, RFID data, order history, menu updates, and transaction logs.
- **Spring Boot REST APIs**: Provide endpoints for mobile and device clients, handling everything from authentication to data validation.
- **Flutter Mobile Application**: Designed for both users and merchants, enabling food ordering, balance checks, budget planning, and live queue visibility.

User Roles & Interfaces

The system supports three primary user roles:

- **Customers (Students/Staff)**: Can register via fingerprint or RFID, view available food items, place orders, and manage budgets.
- **Merchants**: Can add daily menu items, view and manage orders in real time, and receive payment notifications.
- **Admins**: Oversee system-wide operations, user role approvals, menu audits, and data analytics through a secure dashboard.

Communication & Integration

The backend communicates with hardware using secure APIs. Each device sends structured data packets (JSON) to backend endpoints for operations such as:

- Fingerprint validation
- RFID payment triggers
- Queue data submission
- Order confirmation and status syncing

All sensitive data, including fingerprints and RFID details, are stored in encrypted format in **mariaDB** to ensure data protection and privacy compliance.

Scalability & Maintenance

The modular design of the system ensures easy scalability—new hardware units can be added without changing the architecture. Additionally, each device has remote logging and manual override capabilities for on-site maintenance.

The **Smart Canteen Management System** is not just an automation tool—it is a comprehensive digital infrastructure tailored to meet the unique needs of educational institutions. By combining real-time analytics, biometric/RFID authentication, mobile-first interfaces, and scalable architecture, it sets a new standard in digital canteen operations.

Technology Stack

Flutter for Mobile Application

In implementing the Smart Canteen Management System, **Flutter** was adopted for building the mobile front end. Flutter's ability to create high-performance cross-platform applications from a single codebase ensures rapid development and a consistent user experience across Android and iOS. Its rich widget library and reactive framework allow for intuitive UI design, making the app user-friendly for students, staff, and merchants alike.

Backend Development

- **Spring Boot** was used for developing a robust, modular, and scalable backend system in **Java**. It provides RESTful APIs for communication with the front end and handles business logic, user management, and order processing efficiently.
- **MariaDB** was chosen as the relational database to store user data, transactions, menus, and payment logs due to its performance, scalability, and MySQL compatibility.
- Backend services are **hosted on AWS EC2 instances running Amazon Linux**, providing a secure, high-availability environment with flexible scaling capabilities.

Hardware Components

- **RFID Reader** for card-based identification and payments
- **Fingerprint Sensor (e.g., FPM10A)** for biometric authentication
- **Raspberry Pi 3B** for processing fingerprint and RFID inputs, written in **Python**, and interfacing with the backend through secure API calls
- **ESP32-CAM module** (optional) for visual monitoring or customer interaction
- **16x2 LCD Display** for showing order totals and confirmation messages

Communication and Hosting

- **AWS EC2 (Amazon Linux)** to host backend services
- **HTTP/REST API** for data exchange between Raspberry Pi modules and the Spring Boot backend

Database

- **MariaDB** was selected to efficiently store structured data such as user accounts, orders, transactions, and food menus. It supports complex queries and integrates well with Spring Boot via **JPA/Hibernate**.

User Interfaces

- **Mobile App (Flutter)**
  - Used by **students** and **staff** to:
    - View daily menus
    - Place online orders
    - Track expenses
    - Pay using fingerprint or RFID
    - View live congestion
  - Messaging system

- **Merchant Dashboard (Flutter mobile App)**
  - View and manage daily menus
  - Handle incoming orders
  - Handle budget requests
  - View daily income
  - Register new user

- **Touchscreen Display Interface**
  - Deployed in canteen using **Raspberry Pi**
  - Allows students to log in via RFID, view menu, and place orders directly

Hardware Implementation

The hardware implementation of the **Smart Canteen Management System** followed a meticulous, phased development process focused on precision, testing, and seamless integration. The system comprises various hardware components, including a fingerprint sensor for biometric authentication, an RFID reader for contactless identification, Raspberry Pi 3B for control and processing, a 16x2 LCD for real-time display, and an optional ESP32-CAM module for camera-based features.

Component Testing

Each hardware module underwent extensive individual testing to ensure its functionality, accuracy, and compatibility with the overall system:

- **Fingerprint Sensor** was tested for registration accuracy, response time, and false accept/reject rates.
- **RFID Reader** was tested for successful tag detection, reading range, and response time under various lighting and environmental conditions.
- **LCD Display** was checked for clear visibility and accurate message rendering.
- **Raspberry Pi 3B** served as the primary microcontroller, and its GPIO interfaces were verified to ensure correct communication with peripheral devices.
- **ESP32-CAM (optional)** was tested for live capture functionality and remote streaming performance.

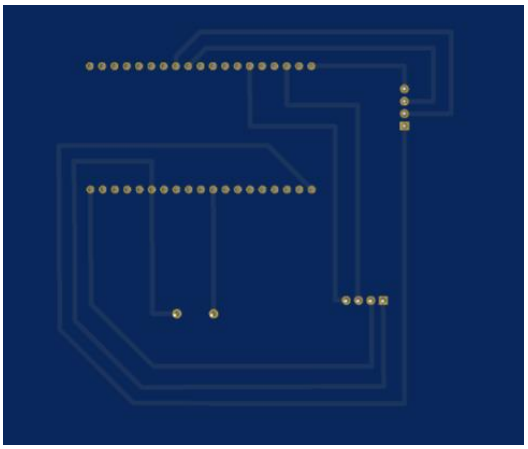
Circuit Development

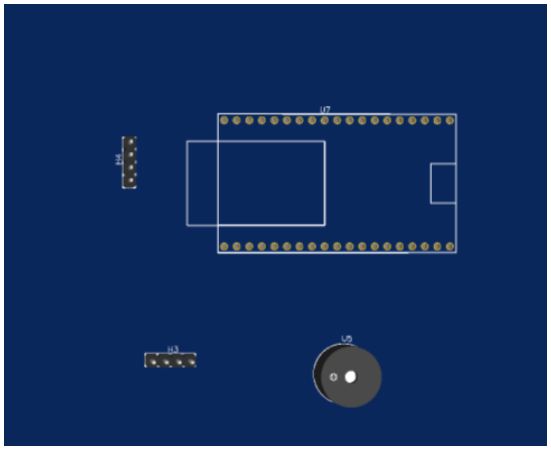
The circuit design was developed incrementally:

- Initial prototyping was conducted on a breadboard for individual modules such as fingerprint, RFID, and display.

- After verifying the stability of each component, integration was carried out into a consolidated circuit on a PCB or perf board

.





Programming in Python

The hardware components interfaced with the **Raspberry Pi** were programmed using **Python**, leveraging libraries such as:

- `pyfingerprint` for fingerprint sensor control
- `MFRC522` for RFID reader interactions
- `RPi.GPIO` for LCD display and I/O handling
- `requests` or `MQTT` libraries for communication with the backend

The use of Python ensured easy maintenance, rapid prototyping, and flexibility to adapt to evolving hardware configurations.

IoT Connectivity via Raspberry Pi

The **Raspberry Pi 3B** functioned as the central hardware controller, equipped with Wi-Fi for seamless IoT connectivity. It communicated directly with the **Spring Boot** backend hosted on **AWS EC2**, sending authenticated user data, order details, and sensor inputs securely over the internet.

Data Transmission to Backend

Once a user authenticated using RFID or fingerprint, the Raspberry Pi transmitted the collected data, including:

- User ID or fingerprint match status
- Selected menu items
- Timestamp of the order

This data was securely sent to the **Spring Boot backend**, which then stored it in the **MariaDB** database and updated the transaction history for both users and merchants.

Display and Confirmation System

A **16x2 LCD display** was used to:

- Show the user's balance
- Display confirmation messages such as "Order Placed" or "Authentication Failed"
- Guide the user through the ordering process step-by-step

Buzzer and Feedback Mechanism

An optional **buzzer** system was incorporated to provide audio feedback for successful or failed actions such as:

- Fingerprint match success
- RFID detection
- Payment success/failure

Mobile Application

The mobile application of the **Smart Canteen Management System** was developed using **Flutter**, a modern UI toolkit developed by Google for building high-performance cross-platform apps. Flutter allows for a consistent, native-like user experience across both **Android** and **iOS** platforms using a **single Dart codebase**.

Flutter was selected for its:

- Cross-platform capability
- Fast development via **hot reload**
- Highly customizable UI components
- Strong performance with native code compilation
- Easy integration with backend APIs and hardware communication systems
- Excellent community support and documentation

User Roles and App Structure

The mobile application is divided into **two primary user groups**:

1. **Users (Students, Parents, Staff):**
   Use the app to authenticate, manage balances, order food, and track expenses.
2. **Merchants (Canteen Staff):**
   Use the app to manage menus, process orders, and track income.

Functionalities of the Mobile Application

User Side

- Register an account with email verification
- Login/logout with secure authentication
- View today's menu and food items
- Place and confirm food orders
- Authenticate via **RFID** or **fingerprint**
- Monitor **total balance** and **expenses**
- View **real-time congestion levels**
- Edit and manage profile information

Merchant Side

- Login/logout
- Add new food items to today's menu
- View current and pending online orders
- Confirm and process orders
- View daily revenue

Flutter Packages Used

The following packages were used to build a responsive and functional app:

- `provider`: state management
- `http`: API requests
- `flutter_secure_storage`: encrypted local storage
- `local_auth`: fingerprint and biometric authentication
- `flutter_local_notifications`: for alerts and reminders
- `fl_chart`: budget and expense visualization
- `dropdown_button2`, `intl`, `connectivity_plus`, `qr_flutter`, etc.

Security Features

- **JWT-based authentication** with secure storage of access tokens
- **Email verification** using secure 6-digit code
- **Biometric login support** for fingerprint authentication
- **Role-based authorization** for users and merchants

API Endpoints

Below are the primary API endpoints used to connect the mobile app to the **Spring Boot backend** hosted on **AWS EC2**:

Authentication & User Management

Function	Method	Endpoint
Register new user	`POST`	`/api/auth/register`
Verify email	`POST`	`/api/auth/verify`
Login	`POST`	`/api/auth/login`
Logout	`POST`	`/api/auth/logout`
Get user profile	`GET`	`/api/user/profile`
Update profile	`PUT`	`/api/user/update`

Fingerprint & RFID

Function	Method	Endpoint
Authenticate via fingerprint	`POST`	`/api/auth/fingerprint`
Authenticate via RFID	`POST`	`/api/auth/rfid`

Menu and Orders

Function	Method	Endpoint
Get today's menu	`GET`	`/api/menu/today`
Place order	`POST`	`/api/order/place`
Get user orders	`GET`	`/api/order/user`
Get merchant orders	`GET`	`/api/order/merchant`
Confirm order pickup	`POST`	`/api/order/confirm`

Budget and Expenses

Function	Method	Endpoint
Set budget	`POST`	`/api/budget/set`
Get current budget status	`GET`	`/api/budget/status`
View expenses	`GET`	`/api/expense/list`

Merchant Menu Management

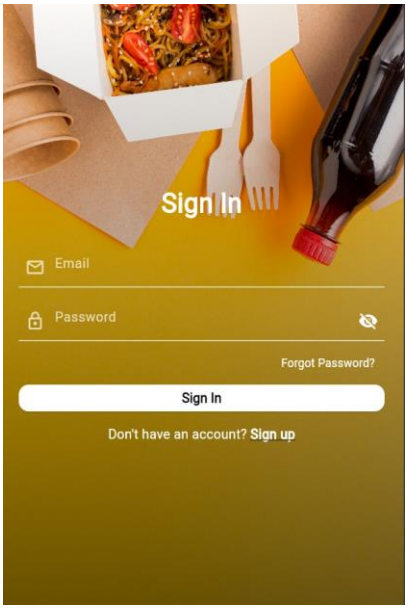
Function	Method	Endpoint
Add food item	`POST`	`/api/merchant/menu/add`
Update food item	`PUT`	`/api/merchant/menu/update`
Delete food item	`DELETE`	`/api/merchant/menu/delete/{id}`
Get menu items	`GET`	`/api/merchant/menu/list`

Notifications (Optional via Firebase or backend trigger)

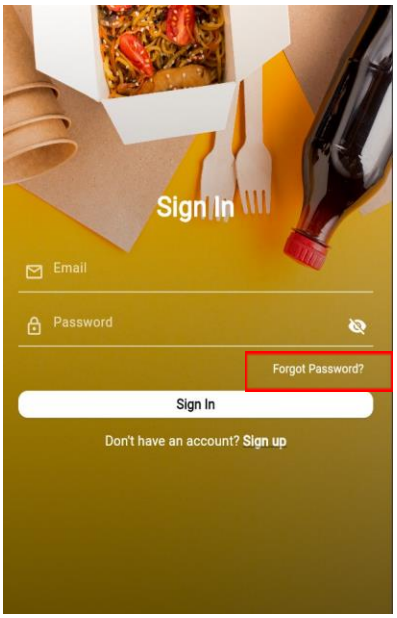
Function	Method	Endpoint
Send notification to user	`POST`	`/api/notify/user`
Send order alert to merchant	`POST`	`/api/notify/merchant`

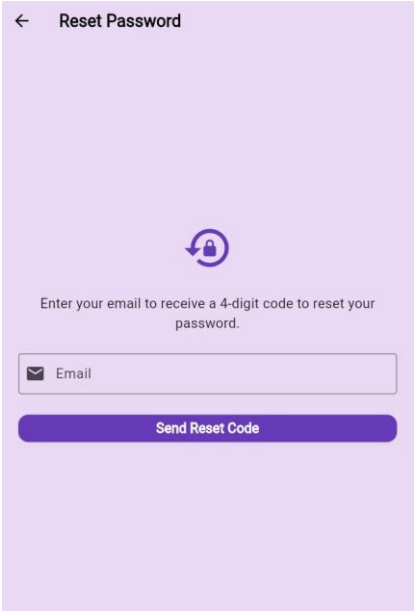
Pages and features

Customer Setup

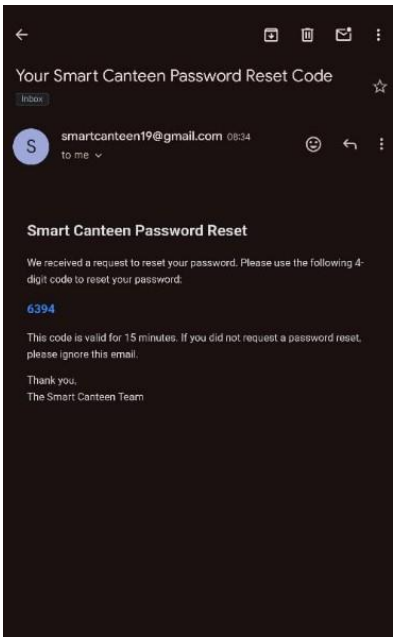


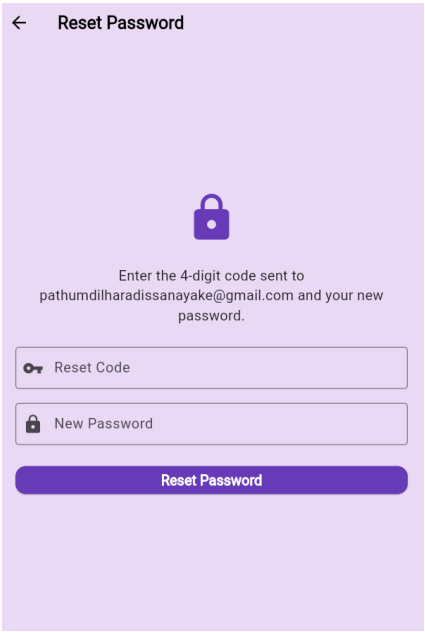
Forgot the password



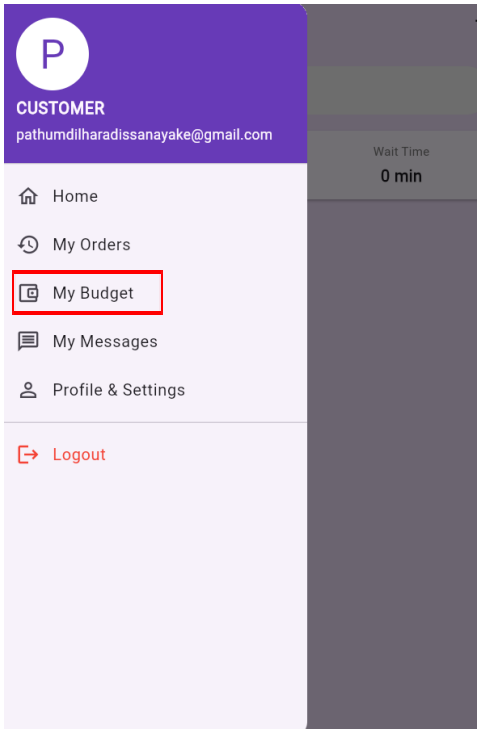


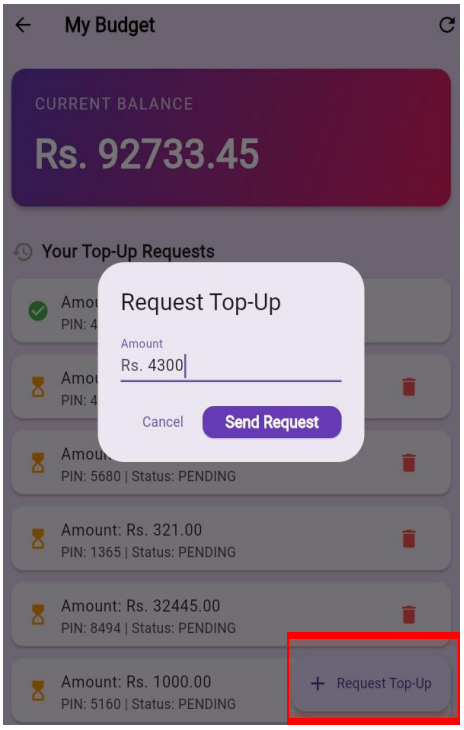
Email verification process



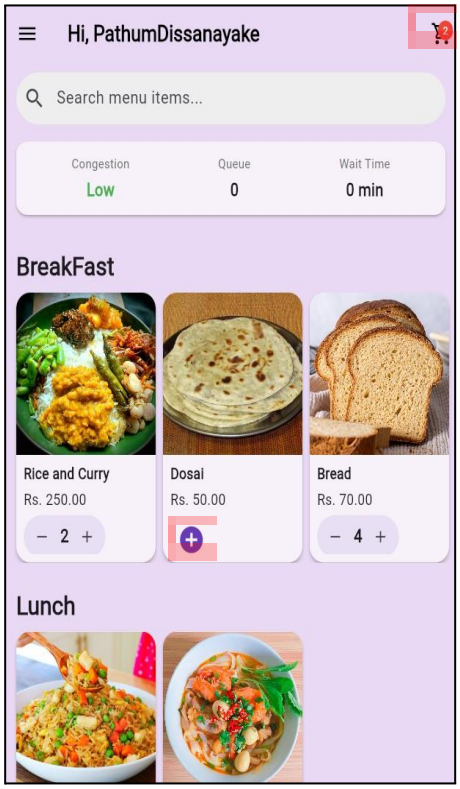


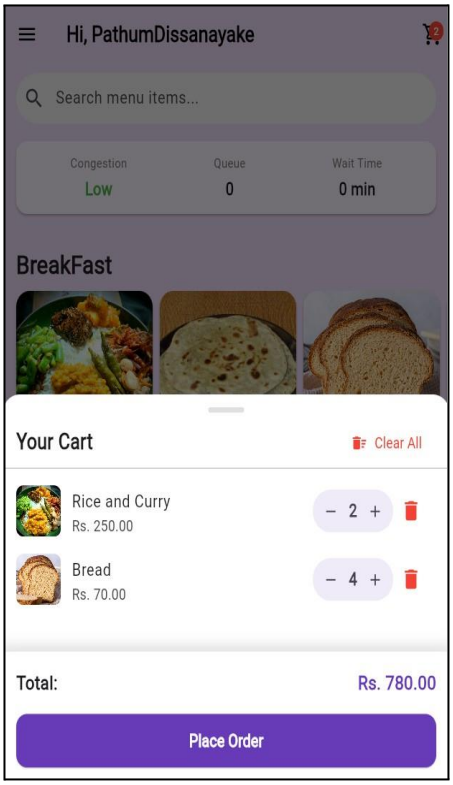
Top up the account





Placing an Order

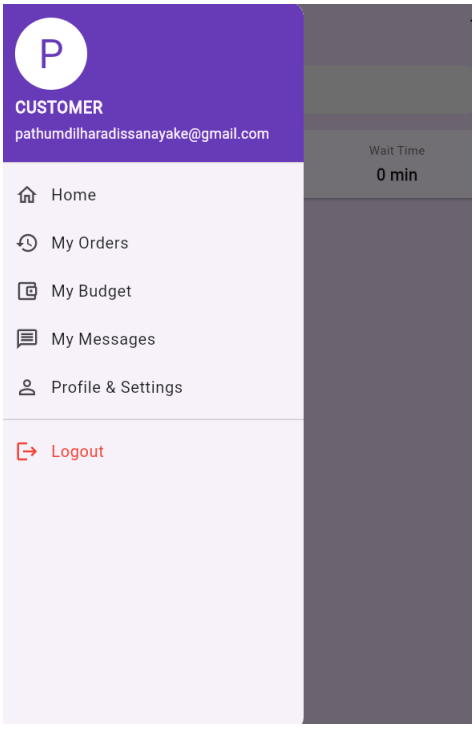


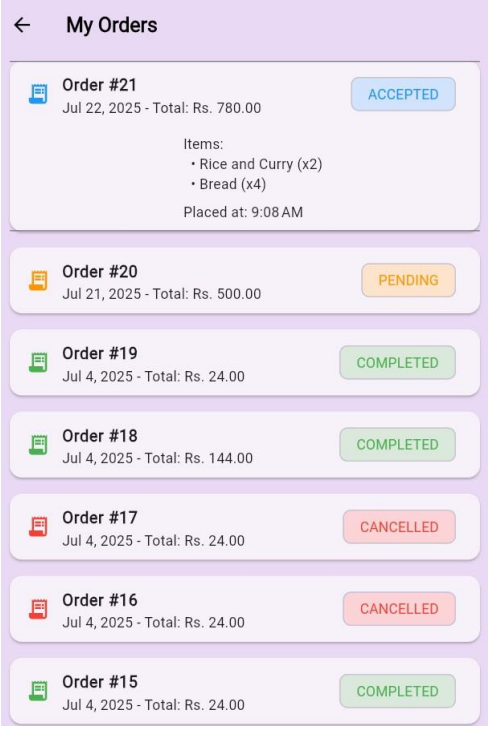


Cancel an Order

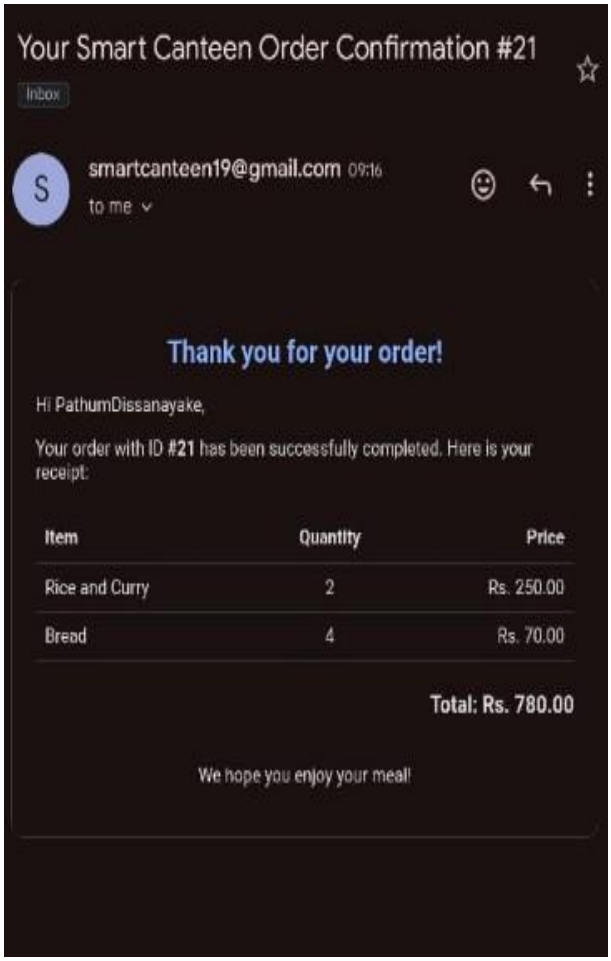
Accept an Order

Get Past Orders

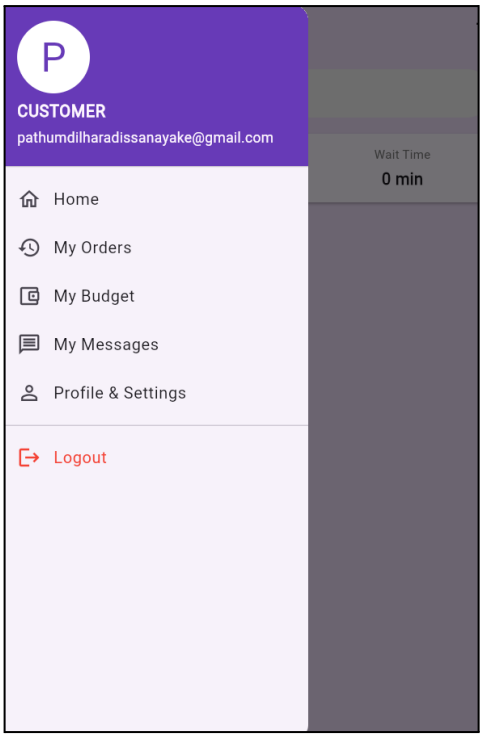


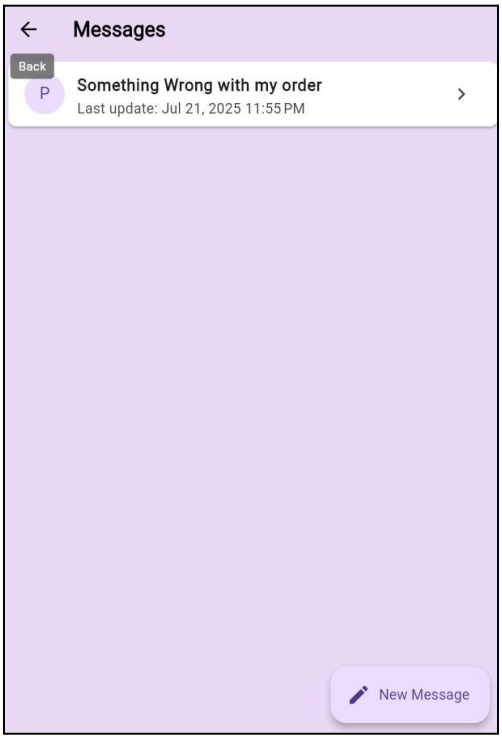


Completion of Order

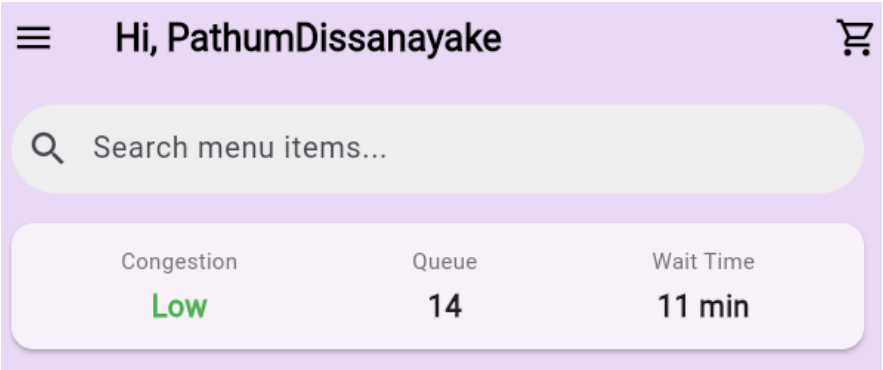


Message Portal

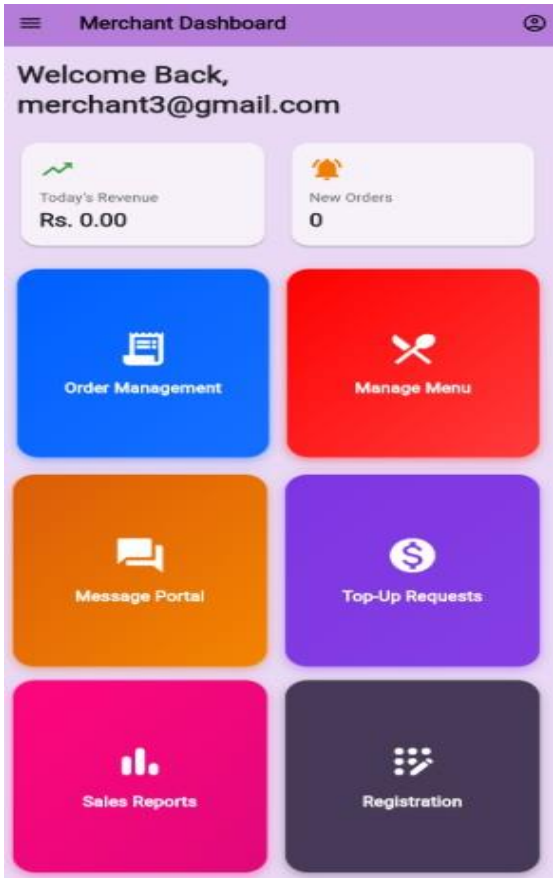


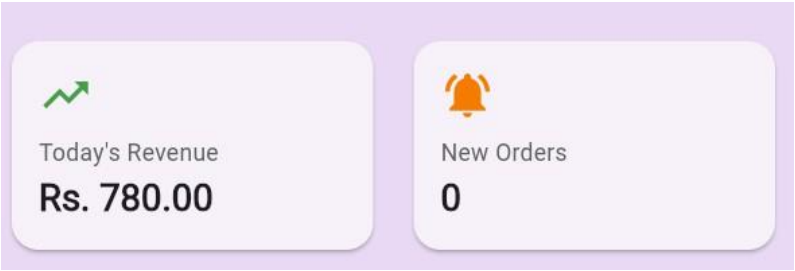


Congestion Alerts

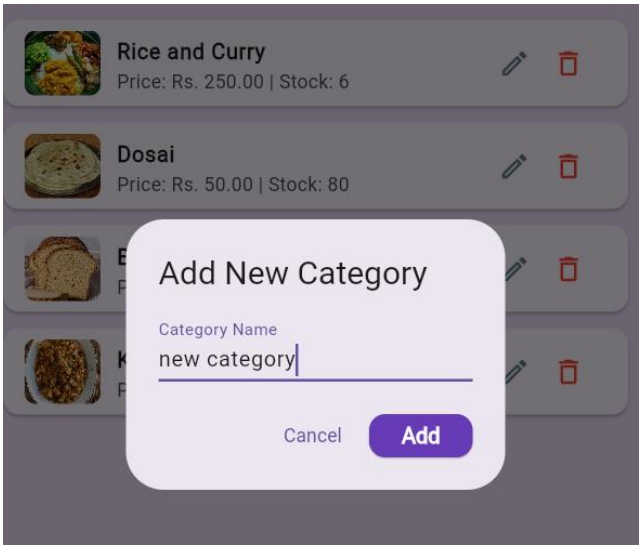


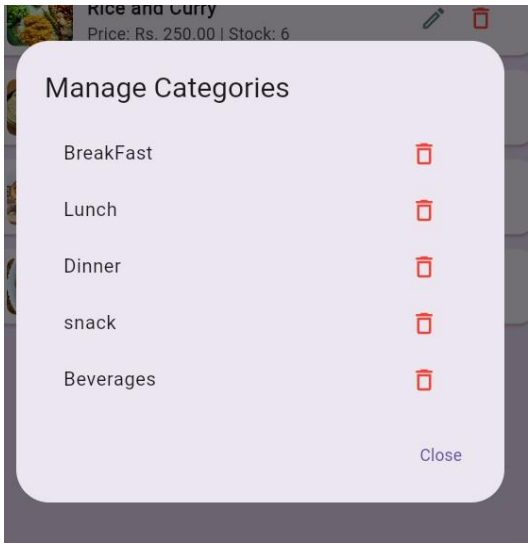
Merchant Setup



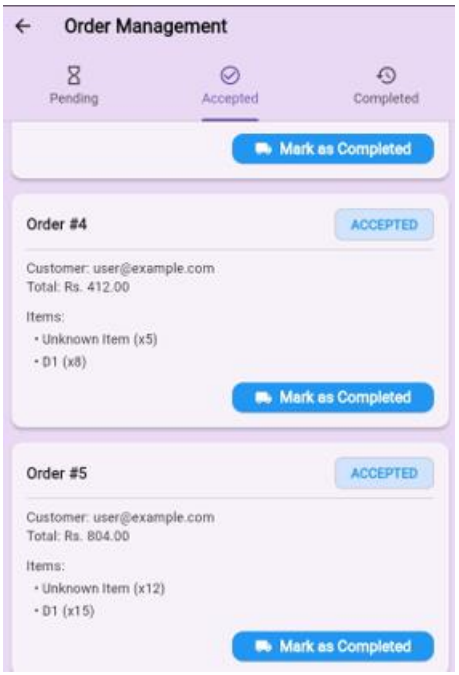


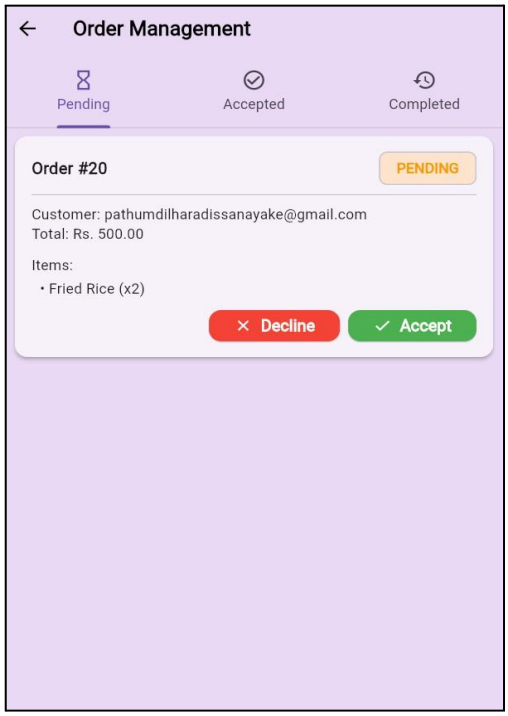
Add Category



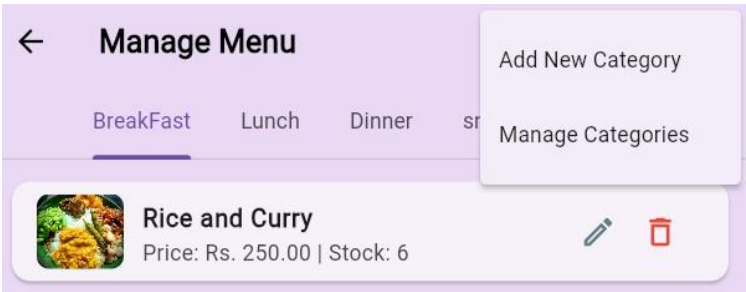


Order Management

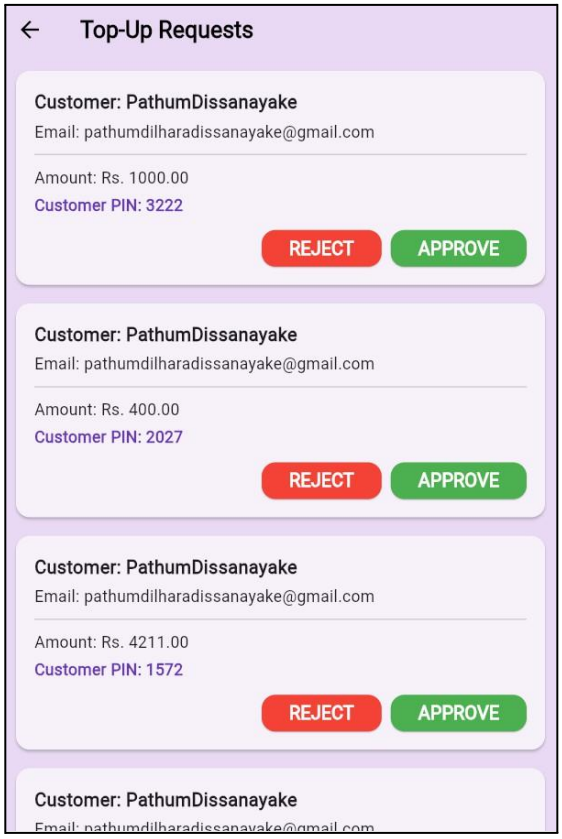




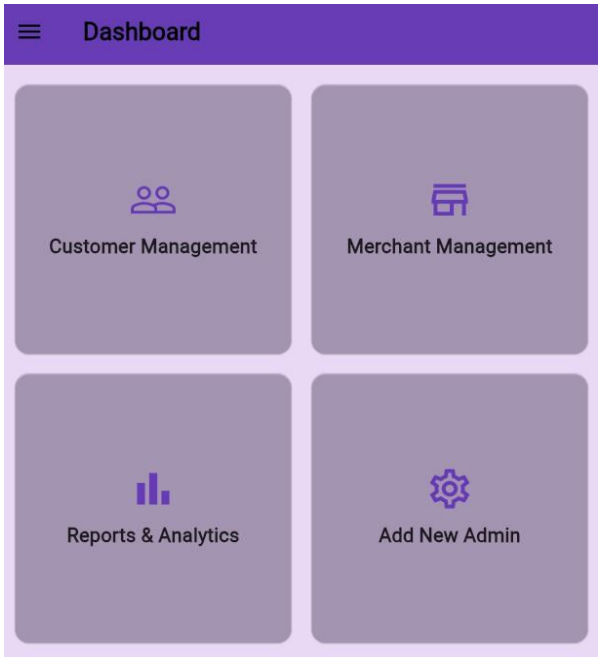
Menu Management



Top-up Request



Admin Functionalities



Database

The heart of our Smart Canteen System is MariaDB , our database. We chose it exactly because it is a rock-solid, stable, and open-source relational database. As an open-source product, it has the support of a vast global community and allows us to be creative without being bound by the technology of one vendor.

MariaDB's biggest strength is how it organizes information. It puts all our information—user profiles and order history, and menu items and payment history—into highly structured tables where every chunk of information has a place. We then create explicit, systematic connections between these tables. For instance, your user account is directly connected to your transaction history, which is linked to the actual menu items you purchased. This cautious, interconnected framework makes certain that our information is always consistent and correct, which is totally crucial when handling finance information and precise stock management.

To communicate with the database, we use SQL (Structured Query Language), industry standard everywhere. It is a powerful language and allows us to perform complex queries with ease, to allow us to ask questions about our information in a very detailed way. We can, for example, generate sales reports for a week, see which dishes sell most on Fridays, or track the use overtime of ingredients. To make these sorts of requests materialize nearly instantly, we have **indexes** on our tables. This operates like an extremely detailed index in a book; rather than looking page by page, the system knows exactly where to locate the information required in less than a second. The outcome is a snappy, responsive system for all users, at all points.

In addition, the database is designed with both survivability and future expansion. We maintain a robust backup and recovery policy that creates periodic, secure backups of all the data so that in case of a rare failure, we can recover all of it very quickly without losing any data. For scalability, MariaDB uses a mechanism called replication. It allows us to create live, synchronized backups of our database across multiple servers. It's like having multiple clerks with the same, minute-to-minute records; when numerous requests arrive at once during a lunch peak, the work is shared between them so that one point isn't congested. That way, our Smart Canteen System doesn't slow down, keeps crashing, or shut down, no matter how busy.

Server

The entire brain of our system the server that intelligently handles all of the nitty-gritty details, manages transactions, and organizes data is meticulously constructed with Spring Boot. We chose this stalwart Java framework because it enables us to create high-performing, secure, and production-ready applications with amazing speed. It deceptively automates much of the annoying setup and config, so our development team can focus on crafting innovative features that directly affect the business and user experience of the canteen. To maintain our system continuously online, always available, and transparently capable of supporting any type of growth in user activity, we smartly deploy our Spring Boot application on an Amazon Web Services (AWS) EC2 instance. Having a global cloud computing leader like AWS allows us to leverage a very reliable, natively secure, and easily scalable server environment.

What this implies is that we can add our computing capabilities like CPU, memory, and storage – dynamically and hassle-free as the popularity of the canteen picks up and usage becomes more intense. This promises a continually smooth and responsive experience for all users, completely removing annoying slowdowns or service interruptions.

Our backend is carefully crafted based on microservices architecture. Instead of a single, monolithic, do-everything application, we've broken down the system into smaller, individual, and specialized services with each task to accomplish a distinct set of functions. For instance, we have one dedicated special service for processing payments effectively, another for securely dealing with user accounts and profiles, and another for dynamically refreshing the menu real-time. This architectural approach brings some significant advantages: it makes the entire system far less likely to fail, since a failure in one service won't bring down the others. It also makes maintenance and development much simpler: if one service needs updating or has a bug fixed, it can be released without affecting the ongoing operation of the rest of the system. To fortify the security of our system, we strictly utilize Spring Security, a feature-rich framework that provides robust authentication and authorization controls, cautiously shielding all sensitive data and strictly preventing unauthorized access.

Security & privacy

In the very center of our Smart Canteen System lies a commitment to keeping your data safe and your experience secure. We've designed a system that protects your information step by step, from food ordering all the way to payment, without ever compromising your privacy. Here's how we do it in plain English:

We start with strong login security measures. Your passwords are securely kept with a method referred to as bycrypt, so even if an intruder breaks into our database, they will be unable to access yours. To make sure that only the right individuals reach the right features, we use something called JWT tokens think of them as secure digital tickets which enable students to order food, employees to accept payments, and admins to deal with the system. For ultra-sensitive operations, like handling financial data, our admins implement two-factor authentication, an added layer of security.

When it comes to your own data, we don't mess around. All data, including your orders or payment details, travels over the internet in industry-leading TLS 1.3 encryption, too high for snoopers. We don't store sensitive payment information on our servers instead, we utilize secure, PCI-DSS-compliant payment gateways to securely process transactions. With items like RFID cards or biometric data (like fingerprints), we use special techniques to encode them into irreversible secure codes, and store only the absolute bare minimum of data for your session. Old data? We regularly sweep it away, so it remains private.

Our biometric and RFID systems are designed with care. Your fingerprint or vein structure is never saved as images equally secure, one-way codes locked in secure hardware. RFID cards have no private data stored, and we require biometric authentication to link them to transactions, so a stolen card can't be traced.

Technically, we've made our system a step ahead of the threats. Menu photos are stored securely in encrypted Amazon S3 storage and can only be accessed with temporary links that expire quickly. We run constant security scans, like penetration tests, to detect and seal holes. Our payment systems are kept on a separate highly secure network to keep them segregated from any illicit access.

Privacy is important to us. You decide if you want to employ features like biometric sign-ins, and we have transparent, uneditable records of your consent. If we measure something like canteen queue times, we only give overall information, nobody's identity is revealed. Your shopping history is not visible, only to you and staff who compile your order. If your account has remained unused for over a year, we automatically delete your data in order to comply with privacy laws like GDPR.

We also rigorously follow rules to keep ourselves compliant with them. We have an internal blockchain system to register important actions like refunds or menu changes so they cannot be altered. Our system is monitored 24/7 for any discrepancies, and we leave it in your hands to handle your data, e.g., to have it erased.

Our security design, or "architecture," is built to keep everything separate and safe. Each element of the system talks to the others in solitude, and we make sure no single breach can destroy the entire. You even get to choose what you share like whether to include dietary practices.

We always hear your feedback and revisit our system to stay secure and safe, so you can enjoy your meals without a care for your information.

Testing

We thoroughly tested every part of our Smart Canteen System for stable operation. Using Postman, we tested all order and payment APIs to ensure they handle different scenarios correctly, e.g., rush lunchtime or invalid input. For real-time functionality like order status updates, we used HTTP test tools to ensure instant notifications work flawlessly.

We thoroughly tested the RFID card system with simulations of hundreds of quick taps for quick recognition. We subjected the fingerprint reader to different finger positions and conditions (e.g., fingers slightly wet) to check for stable recognition. The hardware components (card readers, biometric devices) were stress-tested in real canteen conditions - full house and high temperatures - to check for durability.

We simulated payment breakdowns, network outages, and power loss to ensure the system recovers gracefully without orders being lost. All of the user interfaces were put through intuitive use by first-time users and students. With continuous testing during development and final real-world pilots in university cafeterias, we've built a system that is robust and easy to use on a daily basis.

Maintenance and Updates

For keeping your Smart Canteen System always operating smoothly and securely, we encourage a simple maintenance plan:

1. Software Updates

- We push updates to the app and kiosk software every month with performance improvements and new features like special meals or faster checkout.

2. Hardware Checkups

- We test all equipment every 2 weeks:

- RFID readers (read 100+ cards in a rush)
- Fingerprint scanners (test recognition with different fingers)
- Kitchen printers (print test orders)

3. Security Protection

- Quarterly security audits to ward off new threats
- Immediate updates if any vulnerability is found
- Automatic encryption of all payments and student data

4. Cloud & Database Care

- Daily backup of all orders and accounts
- Monthly server optimization for faster loading
- Storage cleanup of old transaction records

5. User Experience Enhancements

- Monthly menu changes in response to student feedback
- Easy interface adjustments (such as bigger buttons)
- Resolution of reported faults within 24 hours

6. Pre-Launch Testing

All updates are tested:

- Payment failures (simulate rejected cards)
- Network outages (check order recovery)
- Rush hour simulation (100+ orders in 5 minutes)

7. Staff Support

- Training videos for new cashiers
- Printed troubleshooting guides near terminals
- -Dedicated helpline (9 AM-5 PM for instant help)

8.Compliance

- Yearly checks for student privacy regulations
- Automatic removal of unused accounts after 1 year

We maintain all canteen devices over school holidays to prevent disruption. Our maintenance staff attends quarterly for complete system check-overs - testing all scanners, terminals, and connections when classes are in normal operation.

Your strategy to ensure your canteen operates smoothly, transactions are safe, and students enjoy stress-free ordering every time!

Conclusion

Smart Canteen System transforms school lunching into an easy, safe, and modern procedure. Through the substitution of cash queues with lightning-fast RFID taps and fingerprint scanning, we ensure students spend less waiting and more enjoying time.

Our solution brings dependable hardware (card readers, biometric scanners) together with cloud-based software to manage hectic lunch rushes seamlessly. Parents enjoy up-to-the-minute spending alerts, and staff enjoy streamlined operations and automatic tracking of inventory.

We stand by our promise of continuous improvement:

- Frequent updates keep payments secure and introduce new features
- Preventive maintenance avoids disruptions during mealtimes
- Customer feedback directly informs future improvements

As Organizations expand, so does our system – integrating with new meal plans, nutrition trackers, and education programs. The Smart Canteen System is not just about faster lunches; it's about creating a safer, smarter dining experience that grows with your organization’s community.

Appendices

1.Device Care Checklist

- Monthly 15-minute checks for:
- RFID scanners (clean lenses, test 20+ cards)
- Fingerprint readers (test recognition success rate)
- Payment terminals (test dummy transactions)

2. Wireless Updates Guide

Simple 3-step OTA updates:

- Tap "Check Updates" on terminal screen
- Approved update during non-peak hours
- Automatic restart (takes 90 seconds)

3.Security Health Reports

Quarterly summaries showing:

- Vulnerability fixes (e.g., payment encryption upgrades)
- Unusual login attempts
- Data protection improvements

4. System Test Plans

Standard pre-update checks:

- Simulate 200 lunch rush orders in 10 minutes
- Test payment failures/cancellations
- Menu change simulations

5. Interface Change Diary

Version history with screenshots:

6. Training Toolkits

Laminated quick guides for:

- New cashiers (transaction steps)
- Kitchen staff (order alert sounds)
- Admins (meal plan overrides)

7.Compliance Proof

Annual certificates for:

- Student privacy (FERPA/GDPR-K compliant)
- Payment security (PCI DSS Level 4)
- Food safety integrations